BOOKLET

Best Management Practices for Groundnut Cultivation in India

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Vice Chancellor's Message

India is the world's second-largest consumer and number one importer of vegetable oil. Although the oilseed production in India has grown over the years, the production has lagged its consumption, resulting in continuous dependence on import. Groundnut known for its multifarious uses including oil production, direct human consumption as food and also animal consumption in the form of hay, silage and cake. Being a grain legume, peanut has an important nutritional value for human beings, and its nutritional value has been exploited for combating malnutrition in children. Groundnut is third important oilseed crop in India mainly grown in Gujarat, Andhra Pradesh, Tamil Nadu, Rajasthan, Karnataka and Maharashtra. Groundnut has the potential to narrow down the import burden of the vegetable oil as there is huge scope for the improvement in the productivity with the adoption of improved technologies.

The average productivity is 2036 kg/ha in India which is quite low as compared to USA (4400 kg/ha) and China (3710 kg/ha). The low and fluctuating productivity is primarily attributed to cultivation of groundnut is mostly in rainfed condition (80%) especially on marginal lands which are poor in fertility, use of local cultivars, inappropriate plant stand, lower use of fertilizers and manures, poor weed management and higher incidence of disease and insect pest. Many improved technologies have been generated through systemic research on the various aspects of the groundnut however, their adoption taking place at slower rate. Therefore, there is an urgent need to document the innovative groundnut technologies which can help in raising the productivity of groundnut.

Adoption of good agricultural practices is very important in harnessing the full potential of the crop in addition to appropriate variety and quality seed in order to facilitate a synergistic effect on crop productivity. Although generalized agronomic recommendations are often made, they need to be customized to meet the requirements of specific fields and be compatible with the socio-economic conditions under which smallholder farmer's function.

I am extremely pleased to share this space to converse with all of you and to present the Booklet "Best Management Practices for Groundnut Cultivation in India". I hope this bulletin will serves as a handy reference guide for the farmers, extension personnel, students of agriculture and researchers to use improved varieties and associated production practices to increase productivity of groundnut in India.

I congratulate the all authors for giving shape to this important bulletin and convey my best wishes to them for grand success of this bulletin "Best Management Practices for Groundnut Cultivation in India".

(Prof. Balraj Singh)

Preface

India is amongst the largest producer and consumer of vegetable oils in the World. India is a global player in edible oil arena, being the second largest importer, and 3rd largest consumer of edible oil as well as fourth largest oilseed producer. Indian vegetable oil economy is the fourth largest in the world next to USA, China and Brazil. Oilseed crops play the second important role in the Indian agricultural economy next to food grains in terms of area and production. On the oilseeds map of the world, India occupies a prominent position, both in regard to acreage and production. The Indian climate is suitable for the cultivation of oilseed crops; therefore, large varieties of oilseeds are cultivated here.

The major oilseeds cultivated in our country are soybean, rapeseed and mustard, groundnut sunflower, castor, sesame, niger, safflower and linseed. Three main oilseeds namely, groundnut, soybean, and rapeseed-mustard accounted for over 88% of total oilseeds output. Groundnut is third important oilseed mainly grown in Gujarat, Andhra Pradesh, Tamil Nadu, Rajasthan, Karnataka and Maharashtra with a combined share of about 91% in total groundnut production in the country. Groundnut occupies an area of 21.5% and production of 25.4% out of the total area and production of oilseed crops with an average productivity of 2036 kg/ha in India which is quite low as compared to USA and China. The low and fluctuating productivity is primarily because cultivation of groundnut is mostly done on marginal lands, which are lacking in irrigation and low levels of inputs are used here. Therefore, there is an urgent need to improve the productivity of groundnut with the adoption of improved technologies.

Improved agronomy is crucial in harnessing the full potential of the crop in addition to appropriate variety and quality seed in order to facilitate a synergistic effect on crop productivity. Although generalized agronomic recommendations are often made, they need to be customized to meet the requirements of specific fields and be compatible with the socio-economic conditions under which smallholder farmers function. Information on harvest and post-harvest operations will help ensure that smallholder farmers produce high quality groundnuts fetching them a higher price in the market. Looking to the need of farmers, this publication entitled “Best Management Practices for Groundnut Cultivation in India” provides an overview of different improved technologies to be adopted to enhance the productivity of groundnut. Data and photographs in this bulletin have been taken from the published articles, reports of AICRP on Groundnut, training manuals and various bulletins.

We shall consider our effort fruitful if this Booklet serves the purpose of guiding farmers, extension personnel, and students of agriculture and researchers to use improved varieties and associated production practices to increase productivity of groundnut in India.

We are aware of the possibility of shortcomings and defects in this Booklet. We shall appreciate if the readers could bear with them and communicate the same to us so that we can bring out a better edition next time.

Authors

Introduction

Groundnut also known as peanut, monkeynut, earthnut or moongfali is not only one of the most important oilseed crops but also an important export commodity in India. Globally, Groundnut covers 327 lakh hectares with the production of 539 lakh tonnes with the productivity of 1648 kg per hectare (FAOSTAT, 2021). With annual all-season coverage of, globally, India with acreage of 54.2 lakh hectares ranks first under cultivation and is the second largest producer with 101 lakh tonnes with productivity of 1863 kg per hectare during 2021-22. The cultivation of groundnut in country mainly concentrated to Gujarat, Rajasthan, Tamil Nadu, Andhra Pradesh, Maharashtra, Karnataka, and Madhya Pradesh. Gujarat is the largest producer of groundnut contributing 36 per cent of the total production followed by Rajasthan (17 %), Tamil Nadu (7.5 %). It is one of the important export commodity also and India has exported 668,885 MT of groundnut to various countries of the World during the year 2022-23 which worth of 6735 crore.

Groundnut plays an important role in the dietary requirement of resource poor woman and children. Groundnut kernels are also used for the preparation of food products like chikkis, groundnut milk, butter, curd including different bakery products. Groundnut cake obtained after extraction of oil, is used as valuable organic manure and feeding material for livestock. The peanut haulms contain crude protein 8– 5%, lipids 1–3% and minerals 9–10%. These are used as cattle feed either in fresh or in dried stage or preparing hay or silage. The peanut shells or pod walls which constitute nearly about 25% of total pod weight are used as bedding material for poultry or as mulching material during summer season to reduce the evaporative losses. Shell material is also used as filler material for making mixed fertilizers and as insulation material for buildings or as fuel in boilers.

Improved agronomy is crucial in harnessing the full potential of the crop in addition to appropriate variety and quality seed in order to facilitate a synergistic effect on crop productivity. This farmer friendly booklet provides information on improved cultural practices in groundnut cultivation which will empower smallholder farmer to make his/her own decision on various components of integrated crop management technology. Although generalized agronomic recommendations are often made, they need to be customized to meet the requirements of specific fields and be compatible with the socio-economic conditions under which smallholder farmers function. Information on harvest and post-harvest operations will help ensure that smallholder farmers produce high quality groundnuts fetching them a higher price in the market

Table 1. Year wise acreage, production and productivity of groundnut in India

Year	Area (M ha)	Production (M tonnes)	Productivity (kg/ha)
2015-16	4.60	6.73	1465
2016-17	5.34	7.46	1398
2017-18	4.89	9.25	1893
2018-19	4.73	6.72	1422
2019-20	4.30	9.95	2310
2020-21	5.57	10.21	1831
2021-22	5.42	10.10	1863
2022-23	4.91	10.06	2036

Source: agricoop.nic.in.

Cropping Season/sowing time

Groundnut is a tropical plant that requires a long, warm growing season. A well-distributed rainfall of at least 50 cm during growing season, abundant sunshine and relatively warm temperatures are best suited for its cultivation. Groundnut is cultivated in one or more (*kharif*, *rabi* and summer) seasons, but nearly 85% of acreage and production comes from *kharif* crop (June- November). About 85% of groundnut grown in India is sown in the kharif season. *Kharif* groundnut mainly grown in the states of Gujarat, Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh and Uttar Pradesh mostly under rain-fed conditions. The time of sowing varies from June to November depending on the type of soil and rainfall. Groundnut yield in the rainy season is lower than in the post-rainy season due to cloudy weather and the presence of diseases and insect pests.

About 15 % of the crop is raised during the rabi/summer season. *Rabi* groundnut is grown from November to April mostly in central, eastern and southern parts of the country. It is mostly grown in rice fallow situations under residual moisture conditions or with limited irrigation facility and in river-bed fields of Orissa, Assam and West Bengal. In areas where irrigation is available or assured, the summer is ideal for groundnut cultivation. The cultivation of groundnut during January to May/June as summer crop mainly concentrated to some parts of Gujarat, Maharashtra Andhra Pradesh, Karnataka, Tamil Nadu and Rajasthan. Good sunshine and high temperature favour pod formation during summer season.

Soil and field preparation

Groundnut can be grown in range of the soils in India such as sandy, sandy loam and heavy black soils. But well-drained, light-textured, loose sandy-loam or sandy clay loam soils with good drainage, having reasonable high calcium and sufficient sulphur, pH 5.5 to 7.0 and medium to high organic matter are most suitable soils for groundnut production. The highly acidic soils (pH <5.5), and highly alkaline soils (pH >7.8) are not suitable for groundnut production. In case the pH is <5.5, lime in appropriate form and quantity should be mixed thoroughly into the soil before land preparation or at the time of land preparation so as to bring it into the optimal range. The rate of application of lime depends upon the type of lime, soil type and depth of application. As a general recommendation, it would require furrow application of 2.0 t/ha of lime to raise the soil pH from 5.0 to 6.5.

Field preparation for groundnut depends on the soil type and onset of monsoon for rainfed crops and the previous crops grown for irrigated crop. The soils are usually ploughed twice with the summer rains or after application of irrigation water for winter/summer sowing followed by two to three harrowing and pulverizing well to obtain a good tilth for optimum germination. Summer ploughing is advantageous to kill weed seeds and hibernating insects and diseases organisms by exposing them to the heat of summer. In terrace and flat lands of high rainfall areas, raised beds with an width of 90 to 120 cm accommodating 3 to 4 rows of 10-15 cm height are to be prepared to avoid water-logging problems.

Seed quality and treatment

Good quality seed of improved varieties suitable for the area is one of the most important factors for increasing productivity. Good quality seed is essential for establishing the required plant population, good development and yield. Purity, good viability, uniformity of seed size, colour, weight and freedom from seed-borne diseases are the chief characteristics of good seed. This can be achieved by the use of certified seeds. Bunch types have non-dormant seeds whereas semi-spreading and spreading have dormant seeds. The non-dormant seeds germinate immediately after maturity. The dormant seeds usually require a resting period of 60 to 75 days to obtain maximum germination. Dormancy can be broken by exposing seeds to ethrel solution of 250 ppm at the time sowing. Groundnut pods for seed purpose are shelled either by hand or by using hand decorticator or by power operated decorticator about a week in advance of sowing. The viability of seed will be lost if shelled long before seeding. After and during the shelling split or damaged, shriveled immature and infected seeds should be removed

and only well-filled seed should be used for sowing. The seed must be tested for their germination and less than 70% is not considered satisfactory as prescribed as seed standards for certification of labeled seed.

Seed should be treated with Carboxin 37.5% + Thiram 37.5 % DS @ 2-3g/kg kernel to control seed borne diseases. The seed can also be treated with *Trichoderma viride* @ 10 g/kg seed or it can be applied @ 10 kg/ha as soil application. In white grub and termite endemic areas to prevent the seed damage from soil insects at initial stages, imidacloprid 600 FS @ 6.5 ml/kg seed can be used. Thereafter, the kernels should also be treated with suitable *Rhizobium* and PSB cultures (3 packets of each).

Sowing methods

The groundnut is generally sown on flat beds. Some of the improved methods to get higher yield over the conventional method are given below:

1. Criss-cross sowing

In this method, total seed lot is divided into two equal halves; first half of the seed is sown in one direction adopting 30 cm spacing between rows and then the remaining half in the perpendicular direction with the same spacing. Criss-cross method can improve yield by about 18% as compared to conventional sowing by maintaining uniform distribution of seed and optimum number of plants/unit area. The seed rate used in this method is same as in conventional method. The only disadvantage in this method is inter-cultivation and earthing up. So, this method is suitable for light soils.

2. Broad-bed and furrow method (BBF)

BBF method is useful in areas having deep vertisol with high rainfall. This system consists of raised beds of 0.90 to 1.2 m width and 15 cm height with two furrows of 30 cm width on either side. Each raised bed would accommodate three to four rows with 30 cm spacing between rows. On an average, 15% higher yield of groundnut has been reported from the medium black soil over the flat bed.

Raised bed sown groundnut

Flat bed sown groundnut

3. Ridge and furrow method

Groundnut is generally sown on flat beds using a seed drill but sowing on either sides of the ridge helps the plants to prevent from come in direct contact with the excess water. Furrows are more advantageous as they increase moisture recharge in the soil by colleting water and simultaneously help in draining away excess water.

Spacing and seed rate

Inadequate crop stand is one of the major factors limiting the productivity of groundnut. Optimum plant population depends on plant type (bunch or spreading), soil, moisture and management practices. Higher plant population reduces the number of branches per plant while lower plant population increases the number of branches/plant. Number of pods/plant increases with increase in number of branches/plant. Seed rate depends upon the size of kernels, test

weight (100-kernel weight), inter-and- intra-row spacing. The most common spacing recommended for bunch type varieties is 30 cm x 10 cm to achieve a plant population of 3.33 lakh/ha The usual seed requirement of bunch type groundnut is 100-110 kg/ha. The most common spacing recommended for semi-spreading and spreading varieties is 40-45 cm x 10 cm or 30 cm x 15 cm to achieve a plant population of 2.22 lakh/ha in *rabi*/summer groundnut. To achieve the optimum plant stand of semi-spreading and spreading varieties by considering their hundred kernel weight and inter-and- intra-row spacing following seed rates should be adequate.

Table 2 Seed rates of spreading and semi spreading groundnut

Hundred kernel weight (g)	Seed rate (Kernel kg/ha)
40-50	100-110
50-60	120-140
60-70	145-160
70-80	165-185
80-90	190-210

Recommended varieties

Improved, trade-preferred cultivar recommended for the location and cropping season should be selected for sowing. It should also have resistance/tolerance to diseases and insect pests prevailing in the locality.

Table 3 Recently released varieties recommended for cultivation in different states of India

State	**Recommended groundnut varieties**
Andhra Pradesh and Telangana	GPBD-4, Kadiri-5, Kadiri-6, GG 16, TCGS 320, TCGS 29, TCGS 341, TPT 25, ICGV-91114, Kadiri-7, Kadiri-8, Kadiri-9, ICGV-00348, AK-265, Kalahasti, Narayani, Prasuna, Abhaya, Greeshma, Vijetha, ICGV-00350, Dharani, K1812, K 1719, Girnar 4, Girnar 5, TCGS 1522
Gujarat	GG-6, GG-7, GG-16, GG-20, TG-26, GJG-HPS-1, Prutha, JL-501, TG 37 A, GJG 31, GJG 9, GJG 22, GJG 17, GG 34, KDG 123, KDG 128, GJG-HPS-2, Girnar 4, Girnar 5 and Padma, VRI 11, VRI 12, GG 39
Kerala	Sneha, Snighda, GG 16, ICGV00348
Maharashtra	AK 159, GPBD 4, Dh 86, TKG-19A, TLG-45, Phule Unap, Ratneshswar, AK-265, AK-303, TAG-24, Ajeya, Dh 257, ICGV-00348, ICGV-03043, KDG 123, KDG 128, K 1319, AK 335, PDKVG 335
Tamil Nadu	JPBD 4, TMV-13, VRI-6, VRI-7, AK-265, Ajeya, ICGV-00348, ICGV-00350, Vijetha, GG-16, Co 4, Co 5, Co 6, Co 7, KDG 128, KDG 123, Girnar 4, Girnar 5, K 1719, K 1812, Dh 256, VRI 9, VRI 10
Karnataka	GPBD 4, AK-265, Ajeya, TDG 39, VG 9816, K 1319, KDG 128, KDG 123, GKVK 5, VRI-7, GG-16, TGLPS-3, Kadiri Haritandhra, Girnar 4, Girnar 5, Dh 256, Dh 257, K 1812
Rajasthan	RG 638, RG-510, RG 559-3, RG 425, RG 382, HNG-10, HNG-69, HNG-123, GG-7, GG 14, Dh 86, GG- 20, TBG-39, TG 37 A, TAG-24, Girnar-2, Pratap Mungphali-1, Pratap Mungphali-2, Pratap Mungphali-3, KDG 128, KDG 123, Girnar 4, Girnar 5, , VRI 11, VRI 12
Madhya Pradesh	JGN-3, JGN-23, AK-159, GG-8, JL 776, J 53

Haryana	ICGS-1, MH-4, Prakash, HNG-10, Mukta, RG 638, RG-510, RG 559-3, HNG-69, HNG-123, GG-7, GG 14, Dh 86, GG- 20, TBG-39, TG 37 A, VRI 12
Punjab	SG-99, M-548, TG 37 A, GG-21, Girnar-2, HNG-10, HNG-69, J 87, RG 638, RG-510, RG 559-3, VRI 12
Odisha	GPBD 4, Dh 86, Smruti, ICGV-91114, TG-51, Vijetha, Girnar-3, TG–38 B, Vasundhara, TG-37A, Devi, Dh 101, Girnar 2, RG 578, JSP 51, KDG 123
West Bengal	TG-51, Vijetha, Girnar-3, TG–38B, Vasundhara, TG-37 A, Dh 101, Girnar 2, RG 578, JSP 51, KDG 123, KDG 160
Uttar Pradesh	TG 37 A, GG-21, Girnar-2, HNG-10, HNG-69, Prakash, Utkarsh, RG 638, RG-510, RG 559-3
NEH region	TG-51, Vijetha, Girnar-3, TG–38B, Vasundhara, GPBD-5

(Source: Pal, K.K., Singh, R. and Singh, A.L. and Patel, H.V. 2013. Handbook of groundnut varieties of India released during 2001-2021, ICAR-Directorate of Groundnut Research, Junagadh- 362 001, Gujarat, India.)

Cropping systems

Crop sequences: In general, groundnut should be rotated with cereals like pearl millet, sorghum, maize, wheat, rice or other minor millets. This will check the build-up of pests, white grubs, nematodes, soil borne diseases, leaf spots and also maintain soil fertility. Some of the promising crop sequences recommended for different states are given below

Intercropping: The most important cereal intercrops grown with groundnut are bajra, sorghum and maize. The other long duration crops suitable for intercropping with groundnut are red gram, cotton, castor, cassava while short duration crops are sesame, sunflower, cowpea, green gram and black gram. Intercropping systems suggested for different states are presented in table 2 along their proportions with the groundnut as base crop.

Table 4. Promising crop sequences recommended for different states involving groundnut

State	Rainfed (Monocropping)-two years	Residual moisture (Double cropping in one year)	Irrigated (Double or triple cropping in one year
Andhra Pradesh	Groundnut-sorghum	Groundnut-bengal Gram	Groundnut-maize
	Groundnut-millet	Groundnut-Safflower	Groundnut-wheat
	Groundnut-tobacco	Groundnut-sesame	Groundnut-onion
Gujarat	Groundnut-sesame	Groundnut-fodder Sorghum	Groundnut-mustard-green gram
		Groundnut-Mustard	Groundnut-wheat-green gram
Karnataka	Groundnut-sorghum	Groundnut-safflower	Groundnut-wheat
			Groundnut-maize
			Groundnut-sunflower
Maharashtra	Groundnut-sorghum	Groundnut-Safflower	Rice-potato-Groundnut
		Groundnut-fodder maize	Groundnut-*Rabi* sorghum
			Groundnut-safflower

Tamil Nadu	Groundnut-sesame	Groundnut-sesame	Rice-rice- groundnut
	Groundnut-cotton		Groundnut-rice-green Gram
			Groundnut-maize
Rajasthan	Groundnut-pearl millet	Groundnut-barley	Groundnut-wheat-green gram
		Groundnut-Mustard	Groundnut-wheat
Madhya Pradesh	Groundnut-sorghum	Groundnut-safflower	Groundnut-wheat/ mustard
Orissa	Groundnut-sorghum/pearl millet	Groundnut-bengal Gram	Groundnut-rice/ragi
		Groundnut-sesame	Groundnut-coriander/cumin

(Source: Jain, N.K., Garhwal, R.S., Meena, H.N. and Patel, H.V. 2013. Technologies to Enhance Groundnut Productivity –Booklet, Directorate of Groundnut Research, Junagadh- 362 001, Gujarat, India.)

Table 5.Some intercropping systems in different states.

State	Intercropping system	Ratio to base crop (groundnut)
Andhra Pradesh	Groundnut + pearl millet	3 : 1
	Groundnut + cowpea	6 : 1
	Groundnut + red gram	6 :1 or 8 : 1 or 10 : 2
	Groundnut + castor	5 : 1 or 7 : 1
Gujarat	Groundnut + sesame	1 : 1
	Groundnut + sunflower	1 : 1
	Groundnut + red gram	3 :1
	Groundnut + castor	1 : 1
Karnataka	Groundnut + red gram	4 : 1
	Groundnut + cotton	3 : 1 or 5 : 1
	Groundnut + sorghum/ragi	6 : 1
Maharashtra	Groundnut + sorghum	4 : 1 or 6 : 2
	Groundnut + red gram	6 : 1 or 10 : 2
Madhya Pradesh	Groundnut + red gram	8 : 2 or 10 : 2
	Groundnut + soybean	4 : 1 or 6 : 1
	Groundnut + sesame	4 : 1
Tamil Nadu	Groundnut + black gram/green gram	6 : 1
	Groundnut + cotton	5: 1
	Groundnut + castor	7 : 1
	Groundnut + sesame	6 : 1
Rajasthan	Groundnut + pearl millet	4 : 1
	Groundnut + sesame	4 : 1

(Source: Jain, N.K., Garhwal, R.S., Meena, H.N. and Patel, H.V. 2013. Technologies to Enhance Groundnut Productivity –Booklet, Directorate of Groundnut Research, Junagadh- 362 001, Gujarat, India.)

Manures and fertilizers

In general, for obtaining high yield of groundnut application of well decomposed farmyard manure @ 10 t/ha at least three to four weeks 21 days before sowing followed by recommended doses of N, P_2O_5 and K_2O through urea, single super phosphate (SSP) and muriate of potash, respectively is recommended depending upon the initial soil fertility status and nature of crop. If SSP is not available, P_2O_5 can be supplied through DAP, 12:32:16 and other complex fertilizers by suitable calculation according RDF.

Table 6 Recommended doses of N, P_2O_5 and K_2O (kg/ha) for different states

State	Situation	N	P_2O_5	K_2O
Andhra Pradesh and Telangana	Rainfed	20	40	20
	Irrigated	30	60	45
Gujarat	Rainfed	12.5-25	25-60	0
	Irrigated	25-37.5	50-70	0
Karnataka	Rainfed	18	46	25
	Irrigated	25	75	25
Madhya Pradesh	Rainfed	20	40	20
Punjab	Irrigated	15	40	25
Rajasthan	Rainfed	15-20	60	30
	Irrigated	15-20	60	30
Maharashtra	Irrigated	20	40	0
Uttar Pradesh	Rainfed	15	30	45
West Bengal	Irrigated	15	30	45
Tamil Nadu	Rainfed	11	22	33
	Irrigated	22	44	66

(Source: Jain, N.K., Garhwal, R.S., Meena, H.N. and Patel, H.V. 2013. Technologies to Enhance Groundnut Productivity –Booklet, Directorate of Groundnut Research, Junagadh- 362 001, Gujarat, India.)

For a rainfed crop, entire doses of nitrogen, phosphorus and potassium should be applied by placement at seeding in the furrow below the seed at sowing. Now a days, it is very easy place fertilizer below the root zone by using seed cum fertilizer drill.

Special requirement of Calcium and Sulphur

These two nutrients are taken up from the pod zone by the pegs and developing pods. Groundnut has the unique characteristic of uptake of Ca and S by the developing pegs and pods. The major function of calcium in groundnut is the improvement in the quality of the nuts as evidenced by thin strong shell besides well-filled pods. Sulphur is directly involved in the biosynthesis of oil and is usually deficient in the groundnut soils. Gypsum (24% Ca, 18.6% S) is the cheapest source for Calcium and Sulphur elements. Higher quantities of gypsum than necessary are applied at early flowering stage to ensure adequate calcium supply in the pod zone. Application of well - powdered gypsum (30 mesh) @ 500 kg/ha at the time sowing is recommended to overcome the deficiency of Ca and S and increase the pod yield stage.

Correction of micronutrient deficiencies

Groundnut is also sensitive to the deficiency of micronutrients such as B, Zn and Fe. The micronutrient deficiency can be corrected by applying them either to the soil or in standing crop

Table 7. Schedule for controlling micronutrient deficiencies

Micronutrient	Form and rate of allocation to soil	Spray schedule
Boron	Borax 5-20 kg/ha	0.2% borax
Copper	Copper sulphate 5-10 kg/ha	0.1% copper sulphate + 0.05% lime
Manganese	Manganese sulphate 10-50 kg/ha	0.6% $MnSo_4$+0.3% lime
Zinc	Zinc sulphate 10-50 kg/ha	0.5% zinc sulphate + 0.2 % lime
Molybdenum	Ammonium molybdate 0.5-1.0 kg/ha	0.01% ammonium molybdate
Iron	Ferrous sulphate 10 kg/ha	0.5% ferrous sulphate+ 0.05% lime

(Source: Jain, N.K., Garhwal, R.S., Meena, H.N. and Patel, H.V. 2013. Technologies to Enhance Groundnut Productivity –Booklet,Directorate of Groundnut Research, Junagadh- 362 001, Gujarat, India.)

Weed management

Weeds cause much damage to the groundnut crop during the first 35 days of its growth. The most critical period of weed competition is from 3-6 weeks after sowing. The average yield loss due to weeds is about 45%. When once pegging begins (40 DAS), there should not be any disturbance to pegs through manual or mechanical weeding. Important weed flora in the groundnut crop can be categorized in three classes :-

A) **Grassy Weeds:** *Dactyloctenium aegyptium* (Crow foot grass); *Echinochloa colonum* (Water grass); *Panicum repens* (Torpedo grass); *Chloris barbata* (Many spiked chloris); *Commelina benghalensis* (Tropical spider wort); *Cynodon dactylon* (Bermuda grass); *Tragus biflorus* (carrot-seed grass); *Eragrostis ciliaris* (Stink grass); *Brachiaria ramosa* (Signal grass); *Saccharum spontaneum* (Thatch grass); *Sorghum halepense* (Johnson grass).

B) **Broad Leaved Weeds:** *Tridax procumbens* (Coat-button); *Celosia argentea* (Feather cocks comb); *Euphorbia hirta* (Pill-pod euphorbia); *Phyllanthus niruri* (Flyroost leaf flower); *Amaranthus viridis* (Pig weed); *Cleome viscosa* (Spider flower); *Trianthema portulacastrum* (Dessert horse purslane); *Portulaca oleracea* (Common purslane); *Boerhaavia diffusa* (Hog weed); *Tribulus terrestris* (Puncture vine); *Euphorbia pulcherrima* (Common Poinsettia); *Acanthospermum hispidum* (Star bur); *Ageratum conyzoides* (Tropical ageratum); *Croton sparsiflorus* (Croton); *Coccinia grandis* (India ivygourd); *Cassia occidentalis* (Coffee senna); *Digera arvensis* (Digera) and *Solanum nigrum* (Black night shade).

C) **Sedges:** *Cyperus rotundus* (Motha); *Cyperus esculentus* (Yellow nut sedge).

Management

- A good crop cover by adopting right spacing between rows and within the row will smother the weed growth.
- Mulching the soil surface in between rows with crop residue material like straw etc may prevent the germination of weed seeds and at the same time smothers the established young weeds.
- Adopting proper crop rotation practices will help in overcoming the dominance of certain weeds and lessening the weed competition in groundnut crop.
- Intercropping practice in groundnut crop not only provide the monetary and land utilization advantages but also help in smothering the weeds with good crop cover over the land surface.
- Hand pulling of weeds though primitive is an efficient way of controlling weeds especially in groundnut.
- Hand weeding should be done twice, first around 20-25 days after sowing and second at about 35-40 days after sowing.
- Cost effective weed management under rainfed conditions is repeated inter-cultivation (harrowing) up to 35 days after sowing followed by hand weeding.
- Chemical weed control is preferred because of less labour involvement and no mechanical damage to the crop that happens during manual weeding. Moreover, the control is highly effective as the weeds even within the rows are killed, which invariably escape during mechanical control.
- Where labour is scarce and expensive, the use of herbicides either alone or in combination with cultural methods to control the weeds effectively and in time has become a necessity.
- Application of Diclosulam 84% WDG @ 20 to 25 g/ha or Pendimethalin+Imazathapyr 32 EC (Readymix) @1.0 kg ai/ha can be applied to the soil surface within 2 days of groundnut sowing. The effect of pre-emergence herbicides lasts 6-7 weeks after which fresh flushes of weeds emerge.
- To control the weeds at the later stage of crop post-emergence application of Quizalofop ethyl @ 0.50 kg a.i./ha or Imazethapyr @ 0.50 kg a.i./ha should be applied to the crop at 18-20 DAS of groundnut or when weeds are at 3 to 4 leaf stage.
- Integrated weed management involving the above two appears most effective and economical, provided the crop is not subjected to prolonged drought or soil moisture stress during the crop period.
- Application of Diclosulam 84% WDG @ 20 to 25 g ha^{-1} followed by 1 hand weeding at 30-35 DAS or Diclosulam 84 % WDG @ 20-25 g a.i. ha^{-1} (PE) *fb* Quizalofop Ethyl 5% EC 50 g

a.i., ha^{-1} (POE) at 20-30 days after sowing.

Groundnut under weedy check

Groundnut under weed free condition

Water management

Groundnut crop is mostly cultivated during *kharif* under rainfed conditions. Irrigated groundnut accounts for over 20% of the total area under the crop in the country and it yields around 4.2 t/ha.

- There is no necessity for irrigation or rainfall up to 25 days after emergence of groundnut.
- Flowering (20-40 DAS), pod formation and development (40-70 DAS) and pod filling and maturation (70-100 DAS) are most sensitive to soil moisture stress.
- Scheduling irrigation at 25% depletion of available soil moisture (DASM) during moisture sensitive stages and at 50% DASM during other stages results in high pod yield and water use efficiency.
- An IW/CPE ratio of 1.0 at moisture sensitive stages and 0.6 during other stages leads to high water use efficiency. Irrigate at 10-12 days interval is usual practice during summer groundnut.
- At times of deficit supplies, an irrigation at 25 DAS followed by 2 at 15 days interval between 45 and 75 DAS appears to be minimum requirement and it can minimize yield losses due to soil moisture stress. The first irrigation is given at 25 DAS to create moisture stress in the soil which is desirable:
    - to get the good root system.
    - to reduce excessive vegetative growth.
    - encourage the better nodulation
    - induce heavy flowering in a single flush (synchronous flowering)
- Keeping the total quantity of irrigation water applied constant, high frequency irrigation increase the pod field of groundnut on sandy and sandy loam soils.
- Depending on soil type, evapotranspiration and crop duration, water requirement of groundnut ranges between 450 and 650 mm.
- The crop is usually irrigated by check basin method. Border strip is more suitable than other methods.
- Sprinkler irrigation is generally adopted for groundnut considering the crop height and the nature of fruiting. The costs are also low for sprinkler compared to drip.
- The irrigation quantum (mm) is decided after studying the soil texture and its infiltration capacity. Application rate per irrigation should be equivalent to the infiltration rate of the soil of each location approximately this would be 15-20 mm at a time for the soil types mentioned above
- Recently drip irrigation is becoming popular among groundnut growers as it increases crop yield by 25-40% besides better seed quality and saves up to 40-50% irrigation

water compared to flood irrigation.

- For drip irrigation in groundnut, drip laterals (having drippers at 30 cm distance and 1.75-2.0 kg/cm^2 pressure and 4 liter/hour discharge rate) should be installed at the interval of three rows (row spacing 30 cm) or keep 90 cm distance between two laterals which should be operated for 50 minutes in the month of June and 30 minutes during rest of the crop period (July to October) on alternate days. This system saves about 60% water and gives higher yield in comparison to surface irrigation systems.

Overview of groundnut crop under drip irrigation

Plant protection measures

Major insect pest and management

Insect pests cause severe losses to groundnut and are recognized as one of the major constraints in groundnut production. Although more than 90 species of insect pests attack groundnut crop, only a few are economically important over large areas.

White grub

White grubs, *Holotrichia* spp. are serious pests in northern states. The young grubs feed on rootlets and nodules, the older grubs in third instar devour the entire tap root and such plants wither and die. The damage usually occurs in patches in field and the crop nearer to the hosts of adult beetles suffers heavy damage.

Management

- Summer ploughing helps to collect and destroy the larvae and pupae of the pest.
- Early sowing may be useful in escaping from grub damage.
- Setting up light trap especially during May and June months (a day after the first summer rains) to monitor and collect the adult beetles.
- Collect adult beetles of white grubs attracted to trees like neem, Acacia on receipt of monsoon shower and destroy them.
- Branches of neem, tamarind and Acacia trees, around the field may be pruned and sprayed with chlorpyriphos @ 3 ml/lit of water.
- Hand pick the adults resting in the soil during day time and destroy.
- Seed treatment with chlorpyriphos @ 10-15 ml/kg of seed or with imidacloprid 600 FS @ 6.5 ml/kg of seed
- Application of imidacloprid 17.8 SL @ 500 mal/ha or chlorpyriphos @ 4.0 lit of /ha in standing crop.

White grub under field

Adult beetles of white grubs

Tobacco caterpillar, Gram pod borer and Red hairy caterpillar

These defoliators inflict economic losses only when the foliage damage exceeds 25%.

Management

Management can be done by growing of resistant/tolerant cultivar, growing of trap crop such as sunflower and castor bean on borders or in groundnut field (1 plant 20/m), destroying of egg masses on trap crops, encouraging larvae predation by birds by providing perches in the field (10-15/ha) and application of nuclear polyhedrosis virus (NPV) @ 250 LE/ha, obtained from a reliable source and neem seed kernel extract @ extract obtained from 10 kg neem seed powder/ha or application of quinalphos 20 EC @750 ml/ha in standing crop.

Tobacco caterpillar

Red hairy caterpillar

Leafminer

Adult light brown in colour, small blister like mines appear on the upper surface of the leaf, at sever stage, entire leaflet becomes brown and it rolls, shrivels and dries up and gives burnt appearance from the distance in the field.

Management:

Collection and destruction of moths by setting light traps, monitoring of pest using yellow sticky pheromone traps @ 4 per acre. In field condition, if more than 50% of leaf miner infected with natural enemies need not to spray any chemical. Spray with Quinolphos @ 500 ml/ha or Chlorpyriphos @ 1250 ml/ha or Flubendiamide 39.35SC @ 75-100 ml/ha or Spinosad 45 SC @150 ml/ha.

Sucking pests

Aphid

Nymph and adult attack on young branches, suck sap from young leaves and flowers, plants will become weak and stunted, In case of severe infestation, they excrete the honey substance. Flower drop occurs in severe cases.

Jassid

Nymph and adults suck the sap from under surface of the leaves, as result inverted "V" shape yellowing on tip of the leaf. Total leaves become yellowing in severe cases and low temperature and high rainfall favors the pest incidence.

Thrips

Nymphs and adults suck the sap from the leaf, brown color blisters can be seen under side of the leaf, in severe case curling of leaves and stunting of the crop observed. *Thrips palmi* transmits *Groundnut bud necrosis virus* as virus and vector specificity existed.

Aphid

Jassid

Management:

Population of natural enemies *viz;* coccinellids, syrphids and Green lays wings present more than 50% , no sprayings against aphids. Seed treatment with Imidacloprid 600 FS or Thiamethoxam @ 2 ml/kg seed (1:3 ratio chemical and water), border crop with bajra/jowar (4-6 rows), spraying with Imidachloprid @ 150 ml/ha at 20 DAS or Thiaclorpid 480 SC @125 ml/ha or Thiocloprid 25 WG @ 100 g/ha or Acetamiprid 20 SP @ 100 g/ha.

Damage done by thrips

Major diseases and management

Aspergillus seedling blight or collar rot/foot rot

It appears 14 days after sowing. Collar rot led to rapid desiccation of the affected plant, affected tissue at collar region is covered with black mass of spores. In immature plants, lesions develop on the stem just below the soil surface and then spread upward along the branches. In matured plants, symptoms generally do not appear until the wilting of the entire plant is apparent. More prevalent in light sandy soils.

Collor rot Seedling rot

Black conidial head of fungi

Management

Selection of healthy seed, use of tolerant genotypes: J11, JCG-88 and OG-52-1, Seed treatment with Mancozeb 50 %+ Carbendazim 25 % WS @ 3 g/kg seed or Corbaxin 37.5 %+ Thiram 37.5 % DS @ 3 g/kg seed, deep sowing of seed should be avoided, deep ploughing of fields and destruction of plant debris, soil application before sowing with *Trichoderma asperellum* 5 kg/ha after enriched by mixing in 225 kg farm yard manure after 15 days of incubation.

Leaf spot diseases (Early leaf spot and late leaf spot)

Yield loses ranged from 10.0 % to 50.0 %. In the absence of fungicide sprays, leaf spots can cause yield losses >50%. Temperatures in the 25 to 30°C range and high relative humidity (>80.0 %) favors infection and disease development. In case of early leaf spot, the lesions are sub circular and 1-10 mm diameter and dark brown on the upper surface of the leaf. In case of late leaf spot, the lesions on the lower surface of leaf are small, more nearly circular and darker than those of early leaf spot. Both the lesions also appears on the stem, petiole and pegs.

Early leaf spot

Late leaf spot

Management

Seed treatment with Mancozeb 50%+ Carbendazim 25 % WS @ 3 g/kg seed or Corbaxin 37.5 %+ Thiram 37.5 % DS @ 3 g/kg seed, spraying of Mancozeb 75 WP @ 1000 g+Carbendazim 50 WP@ 500 g /ha or Hexaconazole 5 % SC @ 500 ml or Chlorothalonil 75 WP @ 1000 g or Tebuconazole 25.9 EC @ 500 ml/ha in 500 liter of water twice at fortnightly intervals starting from first disease appearance.

Rust Disease (*Puccinia arachidis*)

Orange coloured pustules appear on the lower surface of the leaflets, in severe cases, lesions also appear on other plant parts except flowers

Abaxial Surface (Orange pustules) Adaxial surface

Management

Growing of resistant genotypes viz., Girnar 1, ICGV 87160, ICGV 86590, ICGV 86325, DRG 12, TAG 24, ALR 2, BSR 1, OG-52-1, ALR 3, VRI 5, CO 4, ALR 1, CSMG84-1, ICGS-5, DRG -17 and GPBD4. Removal of infected plant debris, collect seed from disease free areas, seed treatment with Seed treatment with Mancozeb 50 %+ Carbendazim 25 % WS @ 3 g/kg seed or Corbaxin 37.5 %+ Thiram 37.5 % DS @ 3 g/kg seed, spraying of Mancozeb 75 WP @1000 gha or Chlorothalonil 75 WP @1000 g or Hexaconazole 5 SC @ 500 ml or Tebuconazole 25.9 EC @ 500 ml/ha in 500 liter of water at 15 days interval starting from disease appearance.

Stem rot (*Sclerotium rolfsii*)

Appears generally after 70 days of sowing to harvest stage, yellowing and wilting of branches just above the soil, whitish mycelium of the fungus develops around the affected stem above the soil level, infection of pegs and pods occurs in severe case and seeds turn to bluish colour.

Symptoms of stem rot

Management

Follow the deep ploughing in summer, selection of healthy seed, seed treatment with Seed treatment with Mancozeb 50 %+ Carbendazim 25 % WS @ 3 g/kg seed or Corbaxin 37.5 %+ Thiram 37.5 % DS @ 3 g/kg seed, soil application before sowing with *Trichoderma*

asperellum 5 kg/ha after enriched by mixing in 225 kg farm yard manure after 15 days of incubation. Gypsum application @ 500 kg/ ha at the time of pegging stage (45-50 DAS) is recommended. Timely control of foliar diseases lead to reduction of incidence of stem rot at the end.

Dry root rot (*Rhizoctonia bataticola*)

Appears at any stage of the crop and severe during drought period, first water soaked necrotic spots appear on the stem just above the ground level, later develop into blackish when they spread to aerial parts and down into roots. If the infection girdles the stem, wilting follows. Roots are commonly attacked, tap root become black with abundant sclerotia and rotten and shredded. Leaf infection characterized by marginal zonate and irregular spots.

Management

Practice deep ploughing in summer, selection of healthy seed for sowing, seed treatment with Seed treatment with Mancozeb 50 %+ Carbendazim 25 % WS @ 3 g/kg seed or Corbaxin 37.5 % + Thiram 37.5 % DS @ 3 g/kg seed, Soil application of *Trichoderma asperellum* developed by mixing 225 kg FYM in 5 kg *Trichoderma asperellum* and allow to grow for 15 days under shade by covering gunny bags. Timely management of irrigation schedule is required without allowing the plants to dry. Do not allow plants to wilt during critical stages of crop.

Rotten and shredding of tap root

Peanut bud necrosis disease

Initial symptoms appear on young leaflets as chlorotic spots and develop in to chlorotic or necrotic ring spots, terminal bud necrosis on main stem followed by necrosis of top growing buds on all primaries, stunting growth with reduced size of leaflets and petioles, infected plants become dwarf and show axillary shoot proliferation with small leaves having mosaic and distortion of lamina. Early infection leads to no flower and pods. Late infection led to few pods with reduced size and mottled testa.

Initial chlorotic ring spots

Death of top growing bud

Management

Growing border crop of jowar (4 rows) with higher seed rate (@ 200 kg/ha) of groundnut treated with imidacloprid 600 FS @ 1 ml/kg of seed followed by foliar application with thiocloprid 480 SC@ 150 ml/ha at 20 DAS, and acetamiprid 20 SP @ 100 g/ha at 35-40 DAS. Seed treatment with Imidacloprid 600 FS @1 ml/kg seeds + Foliar sprays using Thiocloprid 480 SC @ 0.3 ml/l at 20-25 DAS followed by Fipronil 5SC @ 1 ml/l @ 40DAS and Acetamaprid 20 SP @ 0.2 g/l at 60 DAS).

Harvesting and storage

- As groundnut is an indeterminate crop, hence synchronous maturity of its pods can not be obtained. Therefore, harvesting should be done when 75-80% of pods are fully mature.
- The important indications of maturity are yellowing of foliage, necrotic spotting of leaves, dropping of old leaves, pods become very hard and tough, they give cracking sound when split open with fingers, the inside of the shell turning dark, with netted venation, seed coat develops pink or red colour (normal colour of the varieties) and raising of the soil to the base of the stem is observed.
- Generally harvesting is done by pulling or lifting the plants from the soil with pods intact. If soil moisture is adequate, then hand pulling. If soil is dry, tractor or bullock drawn blades are used for lifting the vines with pods.
- Harvesting before maturity reduces yield and oil percentage and seeds are highly susceptible to aflatoxin. If delayed, results in increased incidence of stem rot, weakening of gynophore/peduncle and some of the pods may remain in soil itself at the time of harvesting.
- The most common method is stripping pods with hand. At the time of harvest, pods usually have moisture content around 40–50% and hence need to be dried to 10% moisture content for safe storage. Drying should be done rapidly to prevent fungal infection. Sun drying is the usual method of drying. Summer groundnut should be dried in shade to prevent loss of viability, if it is for seed purpose.

- Storage at farmer level is invariably in the form of pods. Farmers usually dispose of groundnut pods within a month from drying yard itself. A few store it for 6 months (till *kharif* seeding) in anticipation of high price.
- Pods for seed purpose are stored for 7–8 months. Pods for seed purpose are stored in earthern pots, mud bins or bamboo baskets or Gunny bags having polythene lining.
- If the seed moisture content is above the critical level of 9-10%, then aflatoxin production due to *Aspergillus flavus* just before the post–harvest drying and mould growth at later stage takes place.
- With improved varieties and matching crop production and protection technologies, the productivity of groundnut can be achieved in the tune of 2-3 t/ha for the *kharif* rainfed ecologies and 2.5-4.0 t/ha for *rabi/summer* (irrigated ecologies).

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